TRỊNH XUÂN ĐẠT

Full-Stack & AI Automation Engineer (Junior / Intern) · Remote from Vietnam (UTC+7)

I build AI-assisted products end to end: a payment-integrated e-commerce platform with an AI admin agent, and multi-agent research reported with honest statistics. Final-year IT student at UTH (graduating 05/2027), available for remote work now.

REMOTE · UTC+7 | AVAILABLE NOW | THIRD PRIZE — NATIONAL YOUTH INNOVATION 2026 (AI)

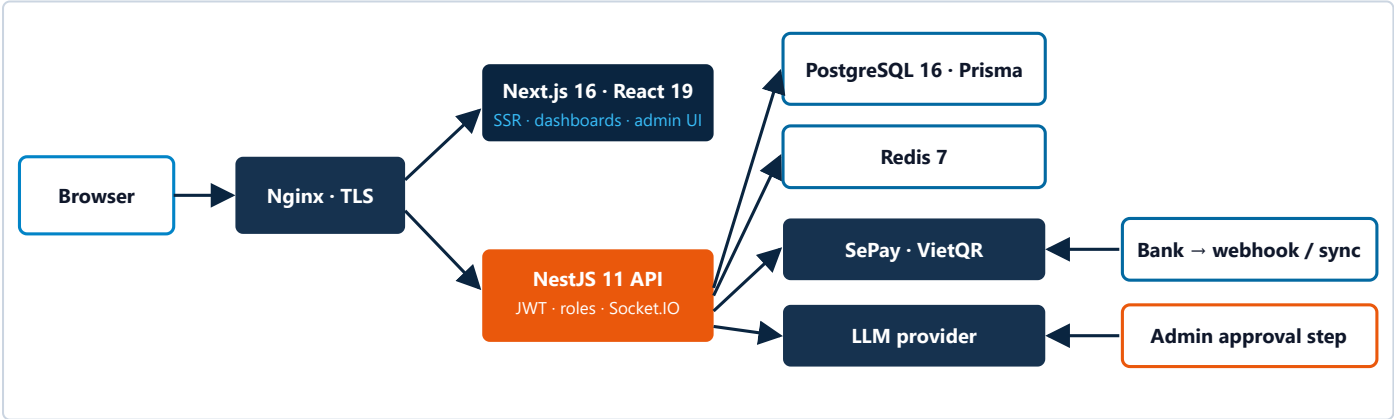
dat02102005@gmail.com | See projects

26 Prisma models · 17 migrations	**22** REST controllers (NestJS)
~48k lines of TypeScript	**30** test spec files
24 containerised services (Traffic MAS)	**5 seeds** 95% CI, Student-t

SIEUTHIMMO — E-COMMERCE PLATFORM WITH AI ADMIN AGENT

FULL-STACK · PRODUCTION

A storefront for digital products and services with wallet, coupons, QR checkout, a customer dashboard, an admin area, realtime support chat, price alerts and AI helpers. Deployed with Docker Compose and Nginx on a Linux VPS. Live: sieuthimmo.tech. Source is in a private repository; I can walk through it on a call.



Payments that credit themselves

- VietQR checkout; incoming bank transfers are matched via SePay webhook/sync and update the order automatically.
- Unique `txHash` constraint on transactions blocks double-crediting at the database level.
- Purchase runs in one DB transaction: balance check, deduction, stock marking, order creation.

AI agent with human approval

- The admin AI never writes directly: it creates a **proposal** with reason, confidence, risk level and a snapshot of the previous state.
- Only allow-listed fields can change; proposals expire after 1 hour and carry a target version; actions land in an audit log.
- LLM replies are validated: upstream error phrases rejected, JSON extracted from mixed text, otherwise a typed error.

Operations & quality

- JWT with refresh sessions, Google OAuth, 2FA, role guards, request logging, timeouts.
- Realtime support chat (Socket.IO) with tickets; semantic product search with keyword fallback if the LLM fails.
- Docs: ADRs, runbook, git workflow, coding convention, release checklist, testing strategy; backup/restore and deploy scripts.

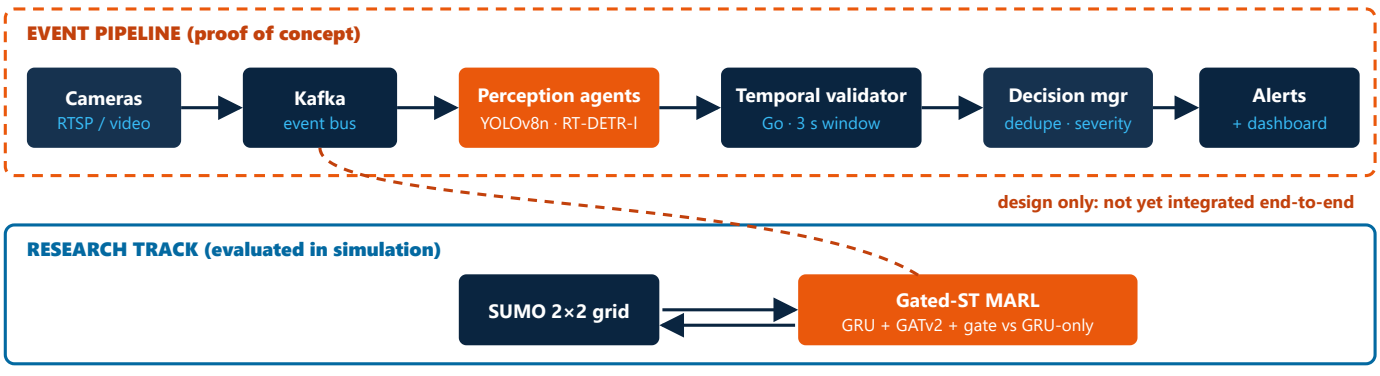
NestJS 11 · Next.js 16 · TypeScript · PostgreSQL + Prisma · React 19 · Tailwind 4 · Redis · Socket.IO · Docker Compose · Nginx · Jest · Webhooks · LLM integration

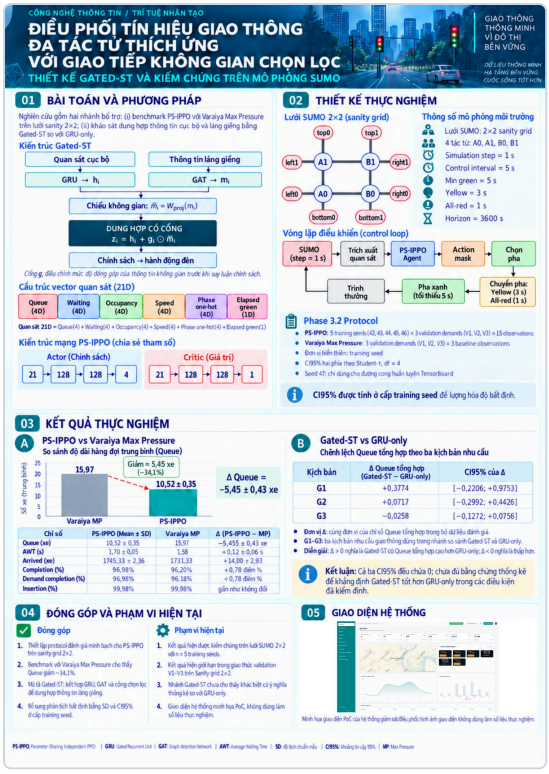
What I would improve next: store money as integers/Decimal instead of floats, move secrets fully out of compose files into a secret store, add end-to-end tests for the payment webhook path, and replace the cookie-session AI provider with an official API key flow.

TRAFFIC MAS — VISION + MULTI-AGENT RL FOR TRAFFIC CONTROL

RESEARCH · EUREKA 2026

Team research project at the UTH AI Lab, submitted to the Eureka 2026 student research awards: (1) a camera → Kafka → validation event pipeline for incident detection, and (2) a multi-agent RL study on a SUMO 2×2 grid comparing a gated spatial-communication model (Gated-ST: GRU + GATv2 + gate) against a matched GRU-only control.



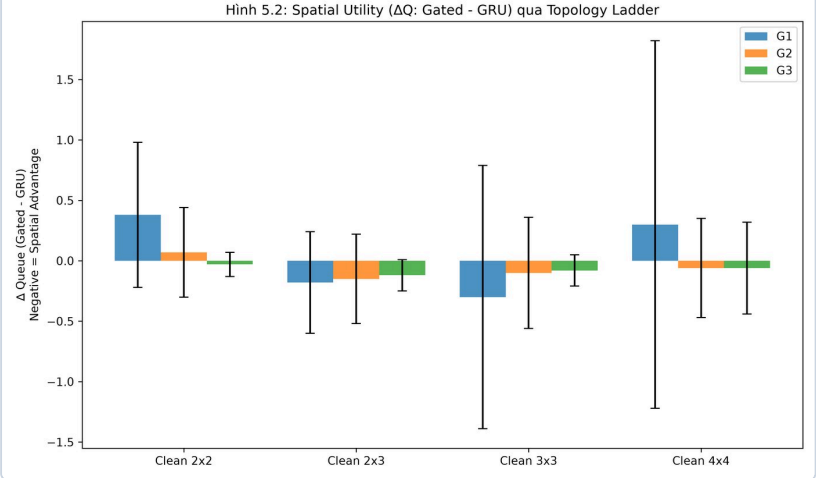


Eureka 2026 poster (Vietnamese): method, SUMO setup, results with 95% CIs, contributions and scope.

Result I can stand behind: Gated-ST vs GRU-only

Scenario	Δ total queue (Gated − GRU)	95% CI
G1	+0.3774	[−0.2206, +0.9753]
G2	+0.0717	[−0.2992, +0.4426]
G3	−0.0258	[−0.1272, +0.0756]

All three intervals contain 0, so there is **not enough statistical evidence** that Gated-ST beats the GRU-only control. 5 training seeds, Student-t intervals, matched control to separate environment shift from communication shift. I report this as a null result rather than overselling it.



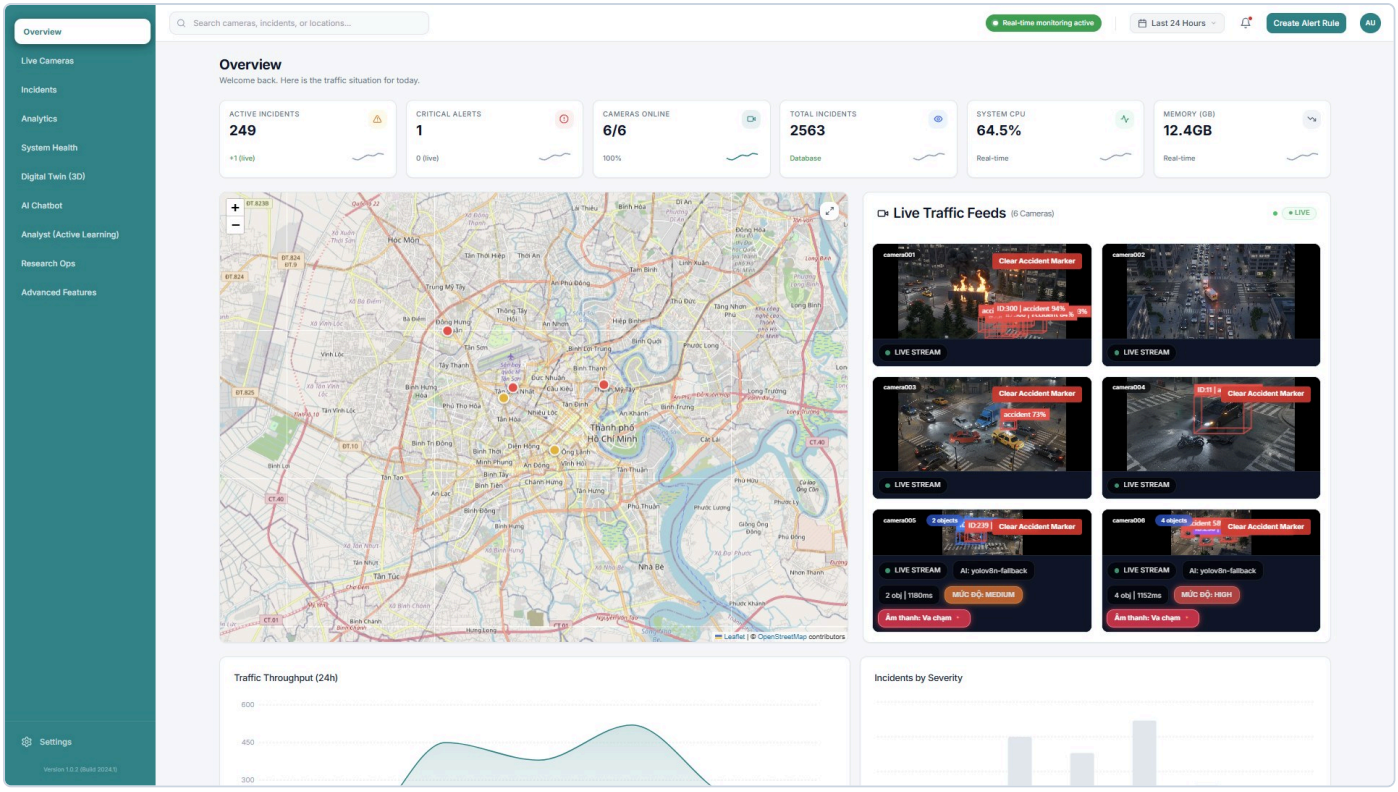
Δ queue across 2×2, 2×3, 3×3, 4×4 grids with error bars; intervals straddle zero.

Engineering, verified

- Measured on CPU: YOLOv8n ≈ **14 FPS**, RT-DETR-l ≈ **1 FPS**.
- 24 services defined in Docker Compose; Kafka and Zookeeper boot verified; Go temporal validator filters events in a 3-second window.
- Reproducibility manifest (git commit, seeds 42–46, CI method) and a validation framework with a 19-requirement traceability matrix, informed by FHWA and ISO 34502.
- 15 unit tests passing for router, validator, incident store and shared core; 34 scenario tests written.

Transparency: scope and limits

- The camera → Kafka → MARL controller chain is a design; it was not run end to end.
- PS-IPPO vs Max Pressure (queue 10.52 vs 15.97 vehicles, about −34%) is from the submitted report; the raw logs were not retained, so I treat it as historical and plan a re-run.
- Throughput figures (e.g. msg/s) and calibration numbers from mock models were removed from the claims.



Operations Dashboard: 6 live CCTV camera streams with real-time AI incident detection (YOLOv8n) & interactive HCMC incident map.



3D Digital Twin: Real-time spatio-temporal intersection simulation with 3D bounding boxes & live queue measurements.

Python · PyTorch · Kafka · Go · YOLOv8 · RT-DETR · GATv2 · PPO / MARL · SUMO · Docker Compose · Next.js dashboard · Prometheus · Grafana · Jaeger

SKILLS, TIED TO THE WORK ABOVE

EVIDENCE-BASED

Area	What I use	Where you can see it
Backend & data	NestJS, Prisma, PostgreSQL, Redis, REST, Socket.IO	SieuthiMMO
Frontend	Next.js (App Router), React, Tailwind, Recharts	SieuthiMMO, Traffic MAS dashboard
LLM & automation	LLM integration with validated outputs, human-approved agent actions, webhooks, scheduled sync, n8n-workflow-aware tooling	SieuthiMMO
ML & research	PyTorch, YOLOv8, RT-DETR, GATv2, PPO, SUMO, Student-t CIs	Traffic MAS
DevOps	Docker Compose, Nginx, Kafka, Prometheus/Grafana, VPS deploy scripts, backups	Both
Ways of working	Written docs, ADRs, runbooks, async communication, reading English technical documents	SieuthiMMO docs

AWARDS

2021 – 2026

10/2026	Third Prize — National Youth Innovation Competition 2026	AI
11/2024	Top 200 Nationwide — Euréka Scientific Research Awards	IT
09/2024	Top 10 Finalists — SSE National Startup Competition	Tech
12/2021	Third Prize (Provincial) — Youth & Children Innovation Contest	Automation

LET'S TALK

REMOTE · UTC+7

Open to remote internships and junior roles in AI automation, LLM-powered products and full-stack development. I work asynchronously, write things down, and can overlap with Asia-Pacific and European hours.

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